

Topics in Sommerfeld Free Electron Theory

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Introduction

This text is an extension of materials learned in an introductory solid-state course for undergraduate physics students (based on the third-year Condensed Matter Physics course at the University of Oxford). We derive an exact expression for the density of states and Fermi energy in an arbitrary dimension, leading to an estimate of the electronic contribution to the heat capacity of metals at a low temperature.

Density of States

We start by defining the density of states $g(k)$ per unit wavevector for a free electronic gas, in a d -dimensional space $\mathbb{R}^d$, as:

$$g(k)dk = (2s + 1) \frac{|d^d \mathbf{k}|}{(2\pi/L)^d} = (2s + 1) \frac{L^d}{(2\pi)^d} \Omega_d k^{d-1} dk \quad (1)$$

where s is the spin of the particle in the gas (for fermions such as electrons, $s = 1/2$), L is the characteristic 1D size of the gas (hence L^d is the characteristic volume in $\mathbb{R}^d$), and Ω_d is the solid angle in $\mathbb{R}^d$ which will be calculated later.

Fermi-Dirac Statistics

Recall that electrons are fermions (overall anti-symmetric), satisfying the Fermi-Dirac statistics, associated with the Fermi factor n_F which specifies the probability of occupying energy E , given as

$$n_F = \frac{1}{e^{\beta(E-\mu)} + 1} \simeq \begin{cases} 1, & E < \mu \\ 0, & E > \mu \end{cases} \quad (2)$$

where μ is the chemical potential, $\beta = 1/(k_B T)$, and $\beta \gg |E - \mu|$ for $T \sim 0$. At low temperature here it takes the shape of a Heaviside function transformed appropriately.

Fermi Energy

The *Fermi energy* E_F is the maximum energy occupied by the electron at $T = 0$ and is also equal to the chemical potential μ here. We estimate E_F by estimating the Fermi wavevector k_F first, since the dispersion relation between energy E and wavevector k is

$$E = \frac{\hbar^2 k^2}{2m} \quad (3)$$

For a total of N particles at very low T , we have $n_F \simeq 1$ as in (2), thus

$$\int_0^{k_F} n_F g(k) dk \simeq \int_0^{k_F} g(k) dk = N \quad (4)$$

Using (1) one can find that

$$(2s+1) \frac{L^d}{(2\pi)^d} \Omega_d \frac{k_F^d}{d} = N$$

$$k_F = 2\pi \left(\frac{nd}{(2s+1)\Omega_d} \right)^{1/d} \quad (5)$$

where $n = N/L^d$ is the number density in $\mathbb{R}^d$. And by the dispersion relation from (3), we get the Fermi energy as

$$\begin{aligned} E_F &= \frac{\hbar^2 k_F^2}{2m} \\ &= \frac{\hbar^2}{2m} (2\pi)^2 \left(\frac{dn}{(2s+1)\Omega_d} \right)^{2/d} \\ &= \boxed{\frac{\hbar^2}{2m} (a_d n)^{2/d}} \end{aligned} \quad (6)$$

where a_d is just a numerical constant given by

$$a_d = \boxed{\frac{(2\pi)^d d}{(2s+1)\Omega_d}} \quad (7)$$

Solid Angle

For the expression (7) the only pain is the solid angle factor Ω_d , which can be elegantly evaluated by considering the property of Gaussian integrals. Obviously, one can write

$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) \\ &= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \end{aligned}$$

$$= \int_0^\infty 2\pi r e^{-r^2} dr$$

More generally

$$\left(\int_{-\infty}^\infty e^{-x^2} dx \right)^d = \int_0^\infty \Omega_d r^{d-1} e^{-r^2} dr \quad (8)$$

The left side of (8) is a standard Gaussian given by

$$\left(\int_{-\infty}^\infty e^{-x^2} dx \right)^d = \pi^{d/2}$$

The right side can be simplified via a substitution $x = r^2$ as

$$\int_0^\infty \Omega_d r^{d-1} e^{-r^2} dr = \frac{\Omega_d}{2} \int_0^\infty x^{d/2-1} e^{-x} dx = \frac{\Omega_d}{2} \Gamma\left(\frac{d}{2}\right)$$

where we used the identity for the Gamma function $\Gamma(t)$ given as

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$$

Hence we equate both sides eventually to get:

$$\begin{aligned} \pi^{d/2} &= \frac{\Omega_d}{2} \Gamma\left(\frac{d}{2}\right) \\ \Omega_d &= \boxed{\frac{2\pi^{d/2}}{\Gamma(d/2)}} \end{aligned} \quad (9)$$

Energy Dependence

Now we return to the density of states. Previously in (1) our density of states is given by $g(k)$ as a function of wavevector k . Now we can simplify the density of states in terms of energy E , by changing the variable as

$$g(E) = g(k) \frac{dk}{dE}$$

Using the dispersion relation in (3) we get

$$g(E) = \frac{1}{2} \left(\frac{2m}{\hbar^2} \right)^{d/2} (2s+1) \frac{L^d}{(2\pi)^d} \Omega_d E^{d/2-1}$$

We now define a density of states per unit volume L^d as

$$D(E) = \frac{g(E)}{L^d} \quad (10)$$

One can simplify $D(E)$ by referring back to (7) and (6) respectively, so we find

$$\begin{aligned}
D(E) &= \frac{1}{2} \left(\frac{2m}{\hbar^2} \right)^{d/2} \left(\frac{(2s+1)\Omega_d}{(2\pi)^d} \right) E^{d/2-1} \\
&= \frac{1}{2} \left(\frac{2m}{\hbar^2} \right)^{d/2} \left(\frac{d}{a_d} \right) E^{d/2-1} \\
&= \frac{1}{2} \left(\frac{2m}{\hbar^2} \right)^{d/2} \left(\left(\frac{dn}{E_F} \right) \left(\frac{\hbar^2}{2m} \right) \right)^{d/2} E^{d/2-1} \\
&= \boxed{\frac{d}{2} \left(\frac{n}{E_F} \right) \left(\frac{E}{E_F} \right)^{d/2-1}} \tag{11}
\end{aligned}$$

At exactly $E = E_F$ we get

$$D(E_F) = \boxed{\frac{d}{2} \left(\frac{n}{E_F} \right)} \tag{12}$$

One can do a sanity check for all the results of (9), (12), (11) and (7) as in *Table 1* below.

Table 1: Properties of Fermi Gas for $d = 1, 2, 3$

d	Ω_d	$D(E_F)$	$D(E)$	a_d
1	2	$n/(2E_F)$	$\propto E^{-1/2}$	$\pi/2$
2	2π	n/E_F	$\propto E^0$	2π
3	4π	$3n/(2E_F)$	$\propto E^{1/2}$	$3\pi^2$

Heat Capacity: Qualitative Estimate

This section is on qualitative analysis only. For a 3D ($d=3$) free electron metal, We can estimate the change in internal energy δU upon an increase of temperature δT . Consider the excitation of particles close to the Fermi surface being proportional to $\Delta N(T) = g(E_F)(k_B T)$, and each one particle gets excited by approximately a factor of $k_B T$. Hence we arrive at

$$\delta U(T) \Delta \sim N(T)(k_B T) \simeq g(E_F)(k_B T)^2 \tag{13}$$

Therefore, the heat capacity C per mole is

$$C = \frac{1}{N/N_A} \frac{\partial}{\partial T} (\delta U) \sim \frac{N_A}{N} g(E_F) k_B^2 T \sim \frac{R}{E_F} (k_B T)$$

Finally

$$C \simeq \xi \frac{R}{E_F} (k_B T) \tag{14}$$

where we have exploited the fact that $k_B = R/N_A$ and $g(E_F) = VD(E_F) \sim N/E_F$. Now we get the famous $C \propto T$ relation for metals at low T , with a constant numerical factor ξ to be

determined. Actually a good fit in experiments gives $C \simeq \gamma T + \alpha T^3$, where, in addition to the Fermi electronic contribution $\sim T$, the additional $\sim T^3$ contribution comes from the oscillation of phonons well-described by the Debye model.